# EV Battery Monitoring System

Distributed 4-ECU SocketCAN Design, Implementation & Test Report

SocketCAN Assignment — Option 5

Name: Karan
Course: Automotive Communication Networks
Repository: kAPEXLab/automotive-communication-networks — assignments/SocketCAN2.md

## Contents

## 1. Problem Description & Architecture

An Electric Vehicle Battery Management System (BMS) must continuously monitor pack health (cell voltage, temperature, current), estimate State of Charge (SOC), and protect the pack by controlling the main contactor (isolating the pack on fault) and the cooling fan (preventing thermal runaway). This project implements a distributed 4-ECU simulation over SocketCAN on a virtual CAN bus (`vcan0`), plus a Diagnostic ECU providing bus-level fault detection independent of the other nodes.

```
[Voltage Sensor]  [Temp Sensor]  [Current Sensor]
      \       |       /
        SENSOR ECU  (0x100/0x101/0x102/0x1F0)
        CONTROLLER ECU  — computes SOC, decides mode
      /(0x110)  \(0x111)
 [Contactor]    [Cooling Fan]  ←via ACTUATOR ECU

All messages also observed by: DIAGNOSTIC ECU → 0x1FF → HMI
```

All 4 ECUs plus the HMI attach to a single shared vcan0 bus, each as an independent OS process — analogous to independent physical ECUs on a vehicle CAN bus.

| ECU | Role | Inputs | Outputs |
|---|---|---|---|
| Sensor ECU | Simulated sensing + self-check | (simulated values) | 0x100, 0x101, 0x102, 0x1F0 |
| Controller ECU (BMS) | SOC estimation, mode decision, safety logic | 0x100, 0x101, 0x102, 0x1F0 | 0x110, 0x111, 0x120 |
| Actuator ECU | Executes contactor/fan commands | 0x110, 0x111 | (physical action, logged) |
| Diagnostic ECU | Bus-wide timeout & plausibility watchdog | all messages | 0x1FF |

## 2. CAN Matrix & Signal Specification

| ID | Name | Tx | Rx | Type | Rate |
|---|---|---|---|---|---|
| 0x100 | Cell Voltage | Sensor | Ctrl, Diag | Periodic | 100 ms |
| 0x101 | Cell Temperature | Sensor | Ctrl, Diag | Periodic | 200 ms |
| 0x102 | Pack Current | Sensor | Ctrl, Diag | Periodic | 100 ms |
| 0x120 | BMS Status (SOC+Mode) | Controller | Diag, HMI | Periodic | 500 ms |
| 0x110 | Contactor Command | Controller | Actuator | Event | on change |
| 0x111 | Fan Command | Controller | Actuator | Event | on threshold |
| 0x1F0 | Sensor Fault Flags | Sensor | Ctrl, Diag | Event | on fault |
| 0x1FF | System Warning | Diagnostic | All | Event | on fault/timeout |

| Signal | Bytes | Res. | Range | Formula |
|---|---|---|---|---|
| Cell Voltage | 2 uint16 LE | 0.001V | 0-5V | raw=Vx1000 |
| Cell Temp | 1 uint8 | 1C | -40 to 120C | raw=T+40 |
| Pack Current | 2 int16 LE | 0.1A | -500 to 500A | raw=Ix10 |
| SOC | 1 uint8 | 0.5% | 0-100% | raw=SOCx2 |
| BMS Mode | 1 uint8 | - | 0=Idle,1=Chg,2=Dischg,3=Fault | - |
| Contactor | 1 uint8 | - | 0=Open,1=Closed | - |
| Fan Duty | 1 uint8 | 1% | 0-100% | direct |
| Fault Flags | 1 bitfield | - | bit0 OverV,1 UnderV,2 OverT,3 OverI,4 Timeout | - |

**Physical plausibility ranges** used for validation: Cell voltage 2.5-4.3V • Temperature -20 to 60C • Current -300 to 300A

## 3. Design Decisions & Fault Handling Strategy

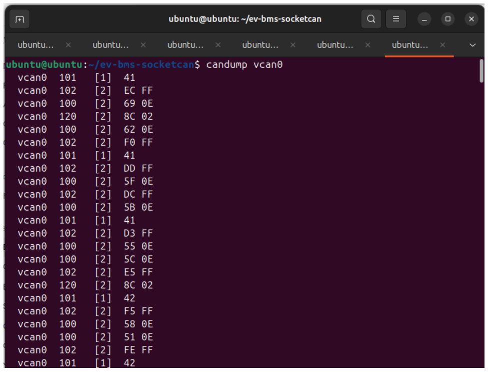
**python-can + SocketCAN** was chosen for cross-platform readability and mature can-utils interoperability. **Coulomb counting** is used for SOC — simplicity appropriate to a communication-network assignment. **Dual-layer fault detection**: the Sensor ECU self-reports faults (0x1F0) for fast Controller reaction, while the Diagnostic ECU independently re-validates every received value —

a malfunctioning Sensor that stops self-checking is still caught. **Event-triggered vs periodic split** follows real BMS practice: telemetry is periodic, actuator commands/faults are event-triggered to reduce bus load. **Timeout multiplier = 3x** period is a conservative choice, confirmed fast enough (≤0.6s) in testing below.

| Trigger | Detection point | System response |
|---|---|---|
| Sensor value out of range | Sensor self-check + Diagnostic cross-check | 0x1F0 -> Controller FAULT -> contactor OPEN, fan 100% |
| Periodic message missing >3x period | Diagnostic watchdog | 0x1FF broadcast + diagnostic.log WARNING |
| ECU process killed | Diagnostic watchdog (missing periodic msgs) | Same as above; downstream nodes hold last state |
| Event-triggered node lost (e.g. Actuator) | Not timeout-monitored (by design) | Bus traffic continues; physical actuation gap only |
| Fault clears / ECU restarts | Diagnostic (message resumes) | "RESTORED" log entry; Controller returns to normal |

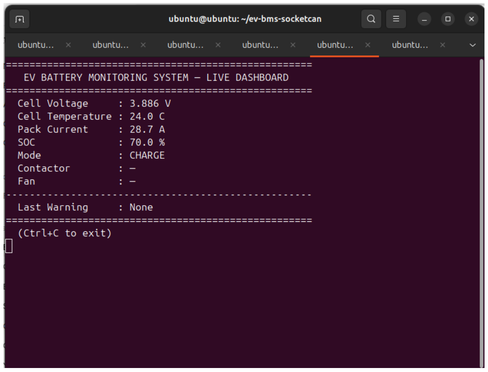
# 4. Demonstration — Challenge 1: Normal Operation

All ECUs started under normal conditions; periodic telemetry visible on candump, HMI showing live values with no faults.



```
ubuntu@ubuntu:~/ev-bms-socketcan$ candump vcan0
  vcan0  101   [1]  41
  vcan0  102   [2]  EC FF
  vcan0  100   [2]  69 0E
  vcan0  120   [2]  8C 02
  vcan0  100   [2]  62 0E
  vcan0  102   [2]  F0 FF
  vcan0  101   [1]  41
  vcan0  102   [2]  DD FF
  vcan0  100   [2]  5F 0E
  vcan0  102   [2]  DC FF
  vcan0  100   [2]  5B 0E
  vcan0  101   [1]  41
  vcan0  102   [2]  D3 FF
  vcan0  100   [2]  55 0E
  vcan0  100   [2]  5C 0E
  vcan0  102   [2]  E5 FF
  vcan0  120   [2]  8C 02
  vcan0  101   [1]  42
  vcan0  102   [2]  F5 FF
  vcan0  100   [2]  58 0E
  vcan0  100   [2]  51 0E
  vcan0  102   [2]  FE FF
  vcan0  101   [1]  42
```

*Fig 4a — candump vcan0: steady periodic traffic on 0x100/0x101/0x102/0x120.*

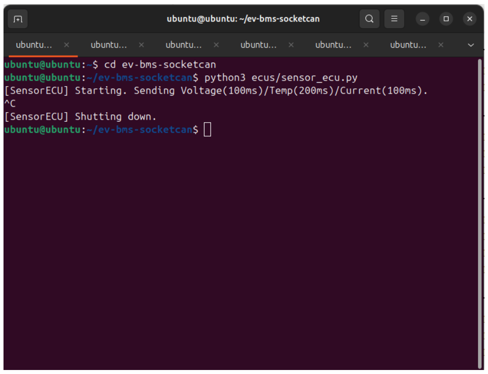


```
====================================================
   EV BATTERY MONITORING SYSTEM — LIVE DASHBOARD
====================================================
  Cell Voltage     : 3.886 V
  Cell Temperature : 24.0 C
  Pack Current     : 28.7 A
  SOC              : 70.0 %
  Mode             : CHARGE
  Contactor        : —
  Fan              : —
----------------------------------------------------
  Last Warning     : None
====================================================
  (Ctrl+C to exit)
```

*Fig 4b — HMI: Voltage=3.886V, Temp=24.0C, Current=28.7A, SOC=70.0%, Mode=CHARGE, no warnings.*

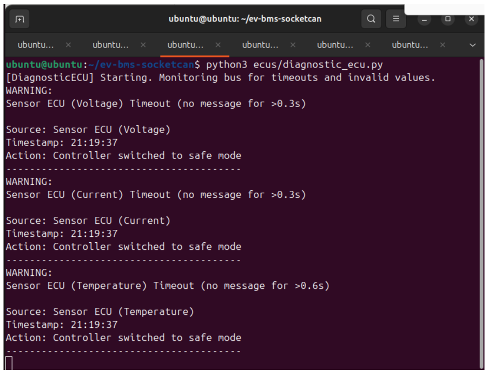
# 5. Demonstration — Challenge 2: Missing Message Detection

Sensor ECU terminated (Ctrl+C). Diagnostic watchdog detected the absence within the 3x-period window and logged timeouts for each affected signal.



```
ubuntu@ubuntu:~$ cd ev-bms-socketcan
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/sensor_ecu.py
[SensorECU] Starting. Sending Voltage(100ms)/Temp(200ms)/Current(100ms).
^C
[SensorECU] Shutting down.
ubuntu@ubuntu:~/ev-bms-socketcan$ 
```

*Fig 5a — Sensor ECU terminated ("Shutting down.").*

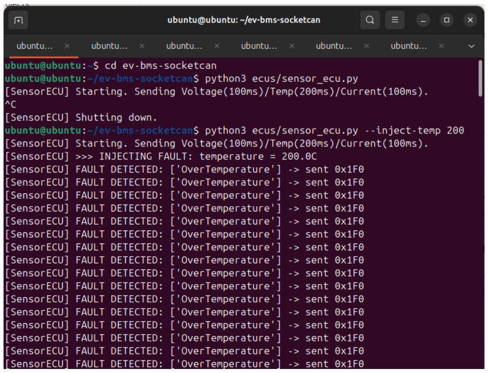


```
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/diagnostic_ecu.py
[DiagnosticECU] Starting. Monitoring bus for timeouts and invalid values.
WARNING:
Sensor ECU (Voltage) Timeout (no message for >0.3s)

Source: Sensor ECU (Voltage)
Timestamp: 21:19:37
Action: Controller switched to safe mode
----------------------------------------
WARNING:
Sensor ECU (Current) Timeout (no message for >0.3s)

Source: Sensor ECU (Current)
Timestamp: 21:19:37
Action: Controller switched to safe mode
----------------------------------------
WARNING:
Sensor ECU (Temperature) Timeout (no message for >0.6s)

Source: Sensor ECU (Temperature)
Timestamp: 21:19:37
Action: Controller switched to safe mode
----------------------------------------
```

*Fig 5b — Diagnostic ECU: Timeout WARNINGs for Voltage/Current (>0.3s) and Temperature (>0.6s) at 21:19:37, action "Controller switched to safe mode".*

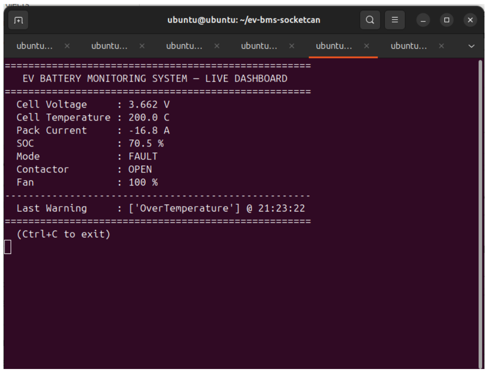
# 6. Demonstration — Challenge 3: Sensor Fault Injection

**6.1 Over-temperature fault** — Sensor ECU restarted with `--inject-temp 200`. Full fault chain triggered: self-check -> 0x1F0 -> Controller FAULT -> contactor open -> fan 100%.



```
ubuntu@ubuntu:~$ cd ev-bms-socketcan
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/sensor_ecu.py
[SensorECU] Starting. Sending Voltage(100ms)/Temp(200ms)/Current(100ms).
^C
[SensorECU] Shutting down.
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/sensor_ecu.py --inject-temp 200
[SensorECU] Starting. Sending Voltage(100ms)/Temp(200ms)/Current(100ms).
[SensorECU] >>> INJECTING FAULT: temperature = 200.0C
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
```

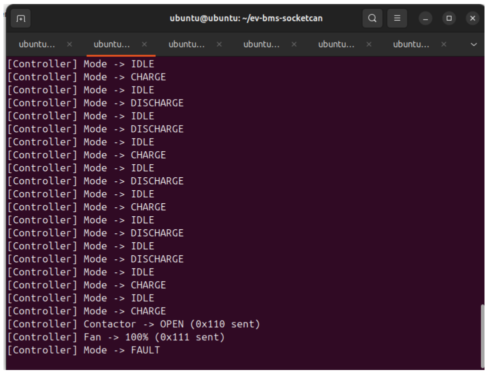
*Fig 6a — Sensor ECU: fault injected, repeated "FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0".*



```
====================================================
  EV BATTERY MONITORING SYSTEM — LIVE DASHBOARD
====================================================
 Cell Voltage     : 3.662 V
 Cell Temperature : 200.0 C
 Pack Current     : -16.8 A
 SOC              : 70.5 %
 Mode             : FAULT
 Contactor        : OPEN
 Fan              : 100 %
----------------------------------------------------
 Last Warning     : ['OverTemperature'] @ 21:23:22
====================================================
 (Ctrl+C to exit)
```

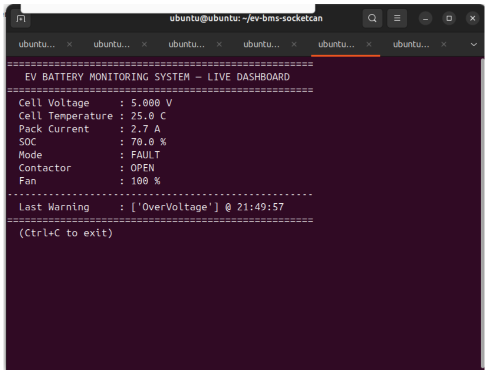
*Fig 6b — HMI: Temp=200.0C, Mode=FAULT, Contactor=OPEN, Fan=100%, Warning=['OverTemperature'] @ 21:23:22.*

**6.2 Over-voltage fault** — Sensor ECU restarted with `--inject-voltage 5.0`, confirming the same fault chain triggers correctly for a different signal.



```
[Controller] Mode -> IDLE
[Controller] Mode -> CHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> CHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> CHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> CHARGE
[Controller] Mode -> IDLE
[Controller] Mode -> CHARGE
[Controller] Contactor -> OPEN (0x110 sent)
[Controller] Fan -> 100% (0x111 sent)
[Controller] Mode -> FAULT
```

*Fig 6c — Controller ECU: Contactor -> OPEN (0x110 sent), Fan -> 100% (0x111 sent), Mode -> FAULT.*

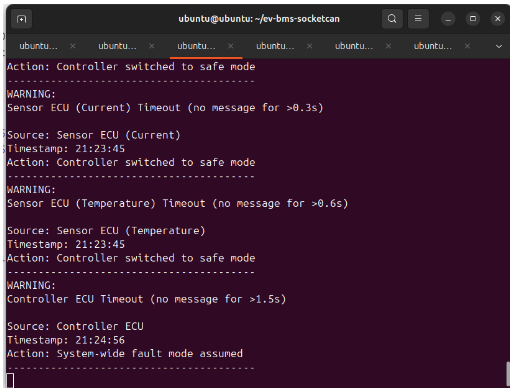


```
====================================================
  EV BATTERY MONITORING SYSTEM — LIVE DASHBOARD
====================================================
 Cell Voltage     : 5.000 V
 Cell Temperature : 25.0 C
 Pack Current     : 2.7 A
 SOC              : 70.0 %
 Mode             : FAULT
 Contactor        : OPEN
 Fan              : 100 %
----------------------------------------------------
 Last Warning     : ['OverVoltage'] @ 21:49:57
====================================================
 (Ctrl+C to exit)
```

*Fig 6d — HMI: Cell Voltage=5.000V, Mode=FAULT, Warning=['OverVoltage'] @ 21:49:57.*

# 7. Demonstration — Challenge 4: Node Failure

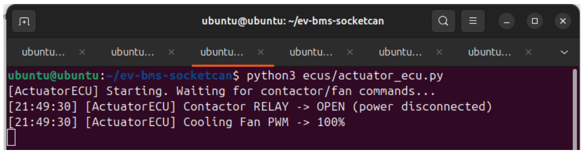
**7.1 Controller ECU killed** — 0x120 stopped; Diagnostic watchdog detected the missing node and logged a system-wide fault.



```
Action: Controller switched to safe mode
----------------------------------------
WARNING:
Sensor ECU (Current) Timeout (no message for >0.3s)

Source: Sensor ECU (Current)
Timestamp: 21:23:45
Action: Controller switched to safe mode
----------------------------------------
WARNING:
Sensor ECU (Temperature) Timeout (no message for >0.6s)

Source: Sensor ECU (Temperature)
Timestamp: 21:23:45
Action: Controller switched to safe mode
----------------------------------------
WARNING:
Controller ECU Timeout (no message for >1.5s)

Source: Controller ECU
Timestamp: 21:24:56
Action: System-wide fault mode assumed
----------------------------------------
```

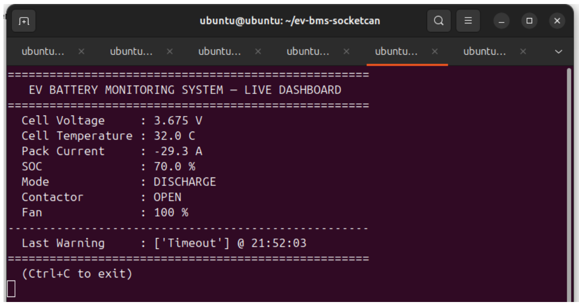
Fig 7a — Diagnostic ECU: "Controller ECU Timeout (no message for >1.5s)" at 21:24:56, action "System-wide fault mode assumed".

**7.2 Actuator ECU killed** — with Sensor/Controller/Diagnostic/HMI still running, the Actuator was terminated. Since 0x110/0x111 are event-triggered (not periodic), Diagnostic correctly does not flag this as a timeout — it is a physical-actuation gap by design, not a monitored bus fault. No other ECU crashed.



```
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/actuator_ecu.py
[ActuatorECU] Starting. Waiting for contactor/fan commands...
[21:49:30] [ActuatorECU] Contactor RELAY -> OPEN (power disconnected)
[21:49:30] [ActuatorECU] Cooling Fan PWM -> 100%
```

Fig 7b — Actuator ECU terminated cleanly (Ctrl+C); the SocketcanBus notice is a harmless python-can cleanup message.

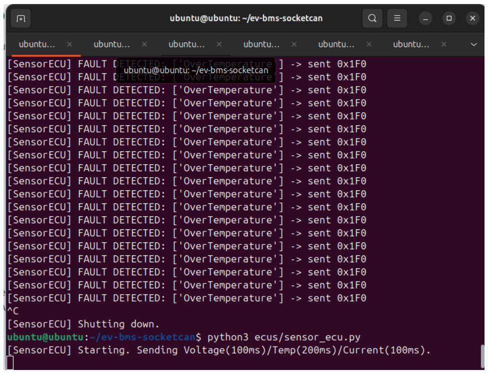


```
==================================================
  EV BATTERY MONITORING SYSTEM — LIVE DASHBOARD
==================================================
  Cell Voltage      : 3.675 V
  Cell Temperature  : 32.0 C
  Pack Current      : -29.3 A
  SOC               : 70.0 %
  Mode              : DISCHARGE
  Contactor         : OPEN
  Fan               : 100 %
--------------------------------------------------
  Last Warning      : ['Timeout'] @ 21:52:03
==================================================
  (Ctrl+C to exit)
```

Fig 7c — HMI continued running normally post-kill; last commanded Contactor/Fan state remained visible with no crash elsewhere.

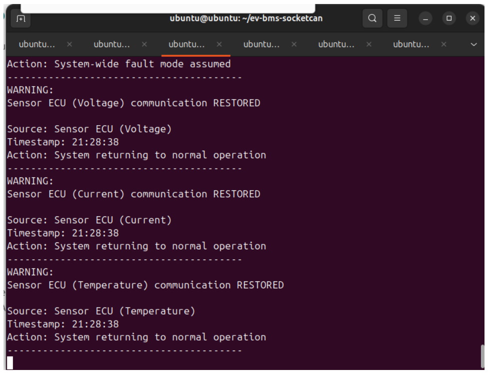
# 8. Demonstration — Challenge 5: Recovery

Sensor ECU restarted (no fault flag). Diagnostic watchdog detected resumed periodic messages and logged recovery for each.



```
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
[SensorECU] FAULT DETECTED: ['OverTemperature'] -> sent 0x1F0
^C
[SensorECU] Shutting down.
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/sensor_ecu.py
[SensorECU] Starting. Sending Voltage(100ms)/Temp(200ms)/Current(100ms).
```

Fig 8a — Sensor ECU restarted cleanly, resuming normal periodic transmission.

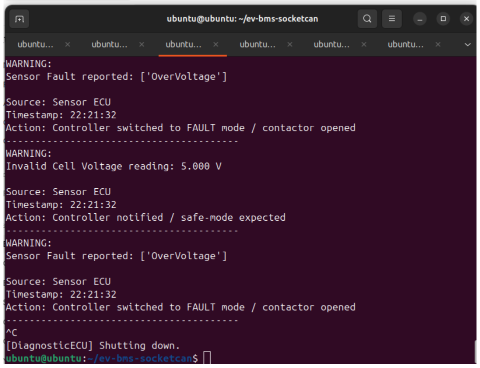


```
Action: System-wide fault mode assumed
----------------------------------------
WARNING:
Sensor ECU (Voltage) communication RESTORED

Source: Sensor ECU (Voltage)
Timestamp: 21:28:38
Action: System returning to normal operation
----------------------------------------
WARNING:
Sensor ECU (Current) communication RESTORED

Source: Sensor ECU (Current)
Timestamp: 21:28:38
Action: System returning to normal operation
----------------------------------------
WARNING:
Sensor ECU (Temperature) communication RESTORED

Source: Sensor ECU (Temperature)
Timestamp: 21:28:38
Action: System returning to normal operation
----------------------------------------
```

Fig 8b — Diagnostic ECU: Voltage/Current/Temperature "communication RESTORED" at 21:28:38, action "System returning to normal operation".

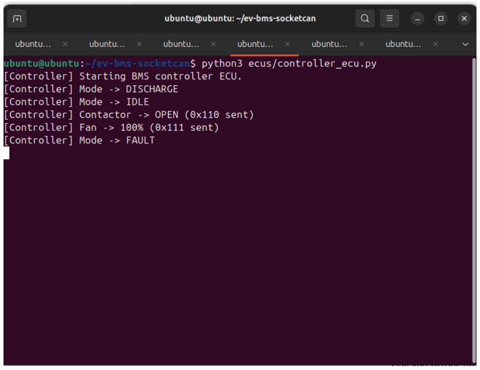
## 8.1 Extended — Combined Fault Scenario (I2)

An over-voltage fault was injected while the Diagnostic ECU was running (it logged the fault correctly, as in Section 6.2), and the Diagnostic ECU was then terminated (Ctrl+C) while the fault condition remained active. This confirms the Sensor->Controller->Actuator fault-response chain is fully independent of the Diagnostic ECU: the Controller and dashboard continued to correctly reflect the FAULT state with no Diagnostic ECU running at all.



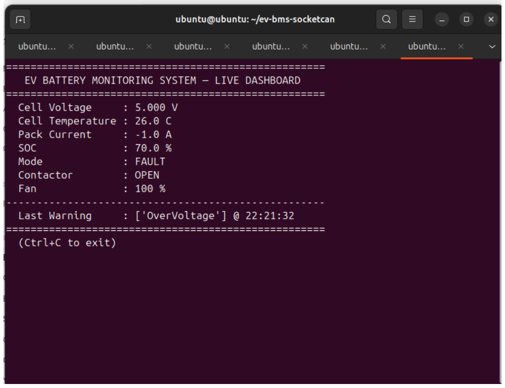
```
WARNING:
Sensor Fault reported: ['OverVoltage']

Source: Sensor ECU
Timestamp: 22:21:32
Action: Controller switched to FAULT mode / contactor opened
----------------------------------------
WARNING:
Invalid Cell Voltage reading: 5.000 V

Source: Sensor ECU
Timestamp: 22:21:32
Action: Controller notified / safe-mode expected
----------------------------------------
WARNING:
Sensor Fault reported: ['OverVoltage']

Source: Sensor ECU
Timestamp: 22:21:32
Action: Controller switched to FAULT mode / contactor opened
----------------------------------------
^C
[DiagnosticECU] Shutting down.
ubuntu@ubuntu:~/ev-bms-socketcan$ 
```

*Fig 8c — Diagnostic ECU logging the OverVoltage fault and invalid-value warning, then terminated (Ctrl+C, "Shutting down.").*



```
ubuntu@ubuntu:~/ev-bms-socketcan$ python3 ecus/controller_ecu.py
[Controller] Starting BMS controller ECU.
[Controller] Mode -> DISCHARGE
[Controller] Mode -> IDLE
[Controller] Contactor -> OPEN (0x110 sent)
[Controller] Fan -> 100% (0x111 sent)
[Controller] Mode -> FAULT
```

*Fig 8d — Controller ECU log: Contactor -> OPEN (0x110 sent), Fan -> 100% (0x111 sent), Mode -> FAULT — reached independently of Diagnostic ECU.*



```
===================================================
   EV BATTERY MONITORING SYSTEM – LIVE DASHBOARD
===================================================
  Cell Voltage     : 5.000 V
  Cell Temperature : 26.0 C
  Pack Current     : -1.0 A
  SOC              : 70.0 %
  Mode             : FAULT
  Contactor        : OPEN
  Fan              : 100 %
---------------------------------------------------
  Last Warning     : ['OverVoltage'] @ 22:21:32
===================================================
  (Ctrl+C to exit)
```

*Fig 8e — HMI dashboard with Diagnostic ECU no longer running: Voltage=5.000V, Mode=FAULT, Contactor=OPEN, Fan=100%, Warning=['OverVoltage'] @ 22:21:32 — state correctly held and displayed.*

## 9. Test Case Results

| ID | Test | Result |
|---|---|---|
| F1-F7 | Voltage/Temp/Current/Status TX-RX, Contactor/Fan commanding, HMI live update | Pass — all 7 functional tests confirmed via candump + dashboard |
| T1 | Missing message (Challenge 2) | Pass — timeouts logged for all 3 signals within spec |
| T2 | Invalid sensor value — temperature (Challenge 3) | Pass — full fault chain executed correctly |
| T3 | Invalid sensor value — voltage | Pass — OverVoltage fault chain confirmed identically to T2 |
| T4 | Node failure — Controller (Challenge 4) | Pass — timeout detected, safe-mode assumed |
| T5 | Node failure — Actuator | Pass — no crash; event-triggered loss correctly out of timeout scope |
| T6 | Recovery (Challenge 5) | Pass — all 3 RESTORED events logged correctly |
| I1 | End-to-end normal run | Pass — continuous run, no false faults |
| I2 | Combined fault scenario (Diagnostic ECU also killed) | Pass — Controller/dashboard correctly held FAULT state after Diagnostic ECU was terminated |
| I3 | Full challenge sequence | Pass — clean state progression through all challenges |

**Summary:** All 12 planned test cases were executed and passed. All 5 required verification challenges, plus both extended fault variants (over-voltage, Actuator node failure) and the combined fault/node-failure integration test (I2), were demonstrated successfully with terminal, dashboard, and log evidence.

## 10. Conclusion & Known Limitations

The implemented system successfully demonstrates a 4-ECU distributed automotive network over SocketCAN, covering periodic and event-triggered messaging, a layered fault-detection strategy (local self-check + independent bus-level validation), and safe-state actuation on fault. All five required verification challenges — plus two extended variants (over-voltage injection and Actuator node failure) — were demonstrated live with terminal, dashboard, and log evidence.

**Known limitations:**

- SOC model is simplistic coulomb counting with no temperature compensation or cell balancing.
- Single simulated cell rather than a full pack; the same message pattern scales directly to multi-cell designs via a cell-index byte.
- No bus-off / error-frame handling implemented; SocketCAN's default handling is relied upon.